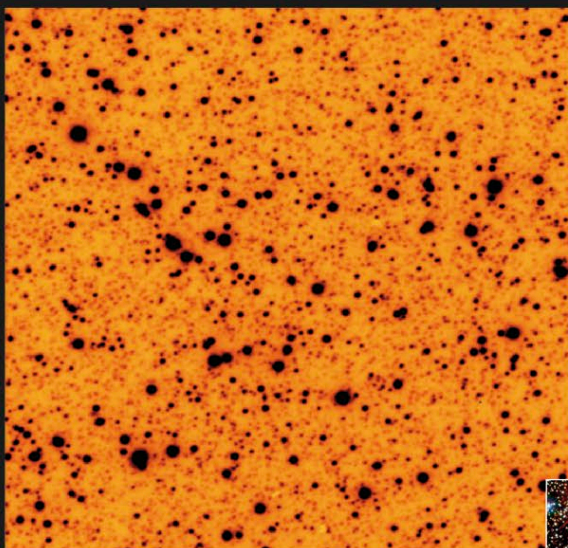


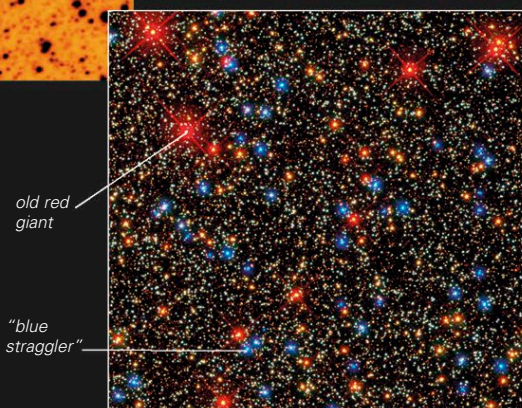
GLOBULAR CLUSTERS

A globular cluster is a massive group of stars bound together by gravity within a spherical volume. Globular clusters can contain between 10,000 and several million stars, all within an area often less than 200 light-years across. As in open clusters, the stars within a globular cluster all have the same origin, and thus similar ages and chemical compositions. Spectroscopic studies of the starlight from globulars reveal that their stars are very old—older than most of the stars currently within the disk of the Milky Way. Analysis of their properties also reveals that they are about the same age, implying that they all formed together over a short period of time. Estimates of their ages vary, but they are thought to be over 10 billion years old. More than 150 globular clusters have been discovered in the Milky Way. Although a few are found in its central bulge, most are located in the halo. The chemistry of globular clusters shows that they represent the remnants of the early stages of the formation of the Milky Way and possibly formed even before the Milky Way had a disk. Four globulars may have originally been part of a dwarf galaxy that has been absorbed into the Milky Way. Globular clusters are made up of Population II stars (see p.227), which have their own independent orbits. These orbits are highly elliptical and can take the globulars out to distances of hundreds of thousands of light-years from the center of the Milky Way. Globular clusters are not unique to the Milky Way, and some galaxies have more globular clusters than our own.



DENSE CLUSTER

An image of part of the Omega (ω) Centauri globular cluster, captured in red light, reveals a great swarm of tightly bound stars. Omega Centauri is one of the densest and most populated globular clusters known within the Milky Way or beyond.



BLUE STRAGGLERS

In the central region of the Omega Centauri cluster, among its old red stars, are seen a few young blue stars. These stars, called "blue stragglers," are thought to have been created by densely packed stars colliding.

CLUSTER EVOLUTION

Star clusters, whether open or globular, are not static through time. Over millions of years, the clusters change physically and the stars within them age and die. However, there are major differences between the evolution of globular clusters and open clusters. An open cluster starts its life with a set of stars of similar chemical composition and age. Over hundreds of millions of years, it loses its members, either due to death of the stars or losing them to the gravitational tugs of other stars within the Milky Way. However, an open cluster can continue to manufacture stars from the original nebulous cloud from which it formed. Because of this, open clusters often contain stars of different ages at various stages of evolution. A globular cluster is more tightly bound and less likely to lose its stars. It also spends most of its time away from the disk of the galaxy, avoiding interactions. In this way, its structure is preserved for thousands of millions of years—far longer than open clusters. Similarly, once a globular cluster has formed, the original gas and dust is ejected and the cluster is then unable to form new stars. As the stars within a globular cluster age and die, so the cluster itself ages and dies.



EVOLVED CLUSTER

At about 1 billion years old, NGC 2266 is a relatively old and well-evolved open cluster. Many of its stars, clearly seen here, have reached the red-giant stage of their life cycle, while young blue stars are also present.

CLUSTER DISTRIBUTION

The difference in the distributions of open and globular clusters within the Milky Way reflects their differences in age and orbit. Open clusters, formed from relatively young, Population I stars, are located within the Milky Way's rotating disk. Globular clusters, made up of Population II stars, have independent orbits mostly located out in the Milky Way's halo.

